JoonSeo Oh

CURRICULUM VITAE

Undergraduate Student / Artificial Intelligence Engineering, Inha University

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RESEARCH INTERESTS

Computer Vision · Human Pose Estimation

Fairness and robustness under challenging imaging conditions.

EDUCATION

Inha University March 2022 – Present

Undergraduate Program in Artificial Intelligence Engineering

South Korea | Expected graduation: August 2027

Cumulative GPA: **4.25 / 4.50**

Freie Universität Berlin Spring 2025

Exchange Student | Germany

RESEARCH EXPERIENCE

Undergraduate Researcher March 2026 – Present

Trustworthy AI Lab, Inha University

Advisor: Prof. Sungho Park

Exposure-Conditioned Fairness in Human Pose Estimation

Ongoing Research | Early Stage

- Working to reproduce UDAPose and evaluate human pose estimation performance on FHIBE under normal- and low-light conditions.
- Investigating whether the performance drop from normal to low light differs across skin-tone groups, focusing on degradation disparities rather than static accuracy differences.
- The broader study aims to characterize skin-tone-specific degradation curves under controlled low-light conditions and explore mitigation if systematic disparities are confirmed.

SELECTED PROJECTS

Automatic Photo Failure Detection Using Classical Machine Learning

Machine Learning Course Project Spring 2026

Individual Project | Code: shot-failure-detection

- Built a three-stage photo quality screening pipeline on KonIQ-10k using 17 handcrafted features, including sharpness, exposure, contrast, noise, and color statistics.
- Compared rule-based classification, logistic regression, random forests, RBF-SVM, and tuned XGBoost; XGBoost achieved a test ROC-AUC of **0.869** and F1-score of **0.823** for MOS-threshold-based quality classification.
- Implemented context-aware heuristics to reconsider images flagged as failures when blur or backlighting may reflect intentional bokeh or silhouette effects.
- Integrated SHAP explanations with rule-based advice to summarize model-attributed quality issues and suggest photographic adjustments.

SELECTED PROJECTS (CONTINUED)

YOLOv5-Based Screening of Handgun Components for 3D-Printing Safety
Creative Design in Artificial Intelligence (AIE1004) Fall 2022

Team Leader | Four-member team

- Led a four-member team developing a prototype to screen 3D-slicer screenshots for handgun components and issue warnings for flagged inputs.
- Fine-tuned a pretrained YOLOv5s detector in PyTorch on a custom three-class dataset (frame, magazine, and slide), using Google Colab.
- Trained on a team-curated dataset of 900 source images combining 450 real-component photographs and 450 Cura screenshots, expanded to 1,652 images through augmentation.

COMPETITION EXPERIENCE

R.O.A.D. Barbados Historic Handwriting Challenge — Zindi
Participant 2026

Historical Handwriting Recognition / Competition Page

- Responsible for image preprocessing and fine-tuning Qwen-family vision-language models for transcription of historical handwritten documents.
- Proposed and experimented with image cropping and color transformations to address preprocessing challenges in degraded document images.
- Contributed to a team approach combining multiple fine-tuned models through ensembling.

HONORS AND SCHOLARSHIPS

Academic Merit Scholarship, Inha University Spring 2026

Exchange Student Scholarship Spring 2025

TECHNICAL SKILLS

Programming	Python
Machine Learning	PyTorch, scikit-learn, XGBoost, OpenCV, NumPy, SHAP, YOLOv5
Tools	Git, GitHub, GitHub Actions, Jekyll, Markdown, LaTeX

LANGUAGES

English: TOEIC 895/990